FURONG JIA

jiafurong2002@gmail.com | Homepage | Google Scholar
Durham, North Carolina, United States

Research Interests

I study **how data shapes LLM behavior** and **how LLMs interpret data**, toward AI that is **reliable, interpretable, and robust**. I examine how **training data** shapes model knowledge and reasoning, how to **evaluate** models faithfully, and how they behave under **distribution shift** across text, time series, and multimodal data. More recently, I study **LLM agents** for scientific discovery and **human–agent interaction**, with applications in **healthcare**.

Education

Duke University — *Durham, NC*
Ph.D. in Computer Science — Aug. 2024 – 2029 (expected)
Advisor: Monica Agrawal — GPA: 4.0/4.0
M.S. in Computer Science (en route) — Sep. 2026

University of Southern California — *Los Angeles, CA*
B.S. in Computer Science; B.S. in Applied and Computational Mathematics — May 2024
W.V.T. Rusch Engineering Honors Program; Viterbi Grand Challenges Scholar — GPA: 3.96/4.0

Research Experience

IBM Research, Core AI Team — *Boston, MA*
Research Intern; Mentor: Akash Srivastava — May 2026 – Aug. 2026

- **When does the harness matter for LLM-driven discovery?** Led a controlled comparison of 11 LLM search harnesses, from best-of-N sampling to evolutionary search and coding agents, on math construction, GPU kernel optimization, and nanoGPT training with Claude Opus 4.6 and Sonnet 4.5, under a shared task interface, evaluators, and API-cost ledger. Choosing a harness often mattered more than upgrading the model, yet the best harness depends on the setup: harnesses differ by over an order of magnitude in API spend per candidate, trading candidate quality against quantity, so rankings change with how the budget is measured and which model is used. Handing the search loop to a coding agent helped with Opus but hurt with Sonnet. [12]

DukeNLP, Duke University — *Durham, NC*
Advisor: Monica Agrawal — Aug. 2024 – Present

- **How faithfully do LLMs acquire probabilistic knowledge?** Continually pretrained OLMo-2 1B and 7B on millions of notes generated from 650 synthetic Bayesian networks, so learned probabilities can be checked exactly. Models get individual probabilities about right but show *relational compression*: they rank outcomes and move predictions in the right direction, yet understate how far apart outcomes are, how much strong evidence should shift a prediction, and how much an unmentioned finding signals absence. Standard metrics miss this. [11]
- **How do LLMs learn clinical knowledge?** Built *MedLingo*, a jargon benchmark from real clinical notes, and counted term–meaning co-occurrences in pretraining corpora. These counts predict which jargon a model understands, yet jargon common in clinical notes is often rare in pretraining data. Models reject disputed medical claims when asked directly but are far likelier to endorse them when the question presupposes them. [4]
- **Can counting match an LLM?** Introduced *FBPR*, a Naive Bayes that ranks MedQA diagnoses using concept–diagnosis co-occurrence counts from an LLM's pretraining corpus (or its open reproduction); it matches that LLM's diagnostic accuracy (OLMo 7B, LLaMA 65B). Yet the two get largely different questions right, so LLMs are not simply aggregating frequencies, and a simple hybrid of the two beats both. [2]
- **Interpreting dataset shift in clinical notes.** Built a benchmark that induces controlled shifts in real MIMIC-IV notes, and *SIReNs*, which explains a shift by showing an LLM representative notes from each dataset. It nearly matches an oracle on clear-cut binary shifts but struggles with gradual continuous ones, which leave only weak traces in each note. Now extending this to LLM agents and harnesses on multifactorial shifts. [3]

Melady Lab, University of Southern California — *Los Angeles, CA*
Advisor: Yan Liu — May 2023 – Jun. 2024

- **GPT4MTS: Prompt-Based Large Language Model for Multimodal Time-Series Forecasting.** Built an LLM pipeline that pairs time series with related news summaries and used it to create a GDELT news-coverage dataset for 55 U.S. regions. Feeding the text to GPT-2 as soft prompts cut MSE by 4% over the best LLM baseline. [5]
- **TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting.** Co-developed and ran training experiments for a GPT-2 forecaster that splits each series into trend, seasonal, and residual parts and uses prompts to adapt to distribution shift. In zero-shot forecasting on unseen datasets, it outperformed baselines evaluated the same way (e.g., 19% lower MAE than PatchTST on ETTm1). [6]

GLAMOR Lab, University of Southern California *Los Angeles, CA*
Advisor: Jesse Thomason Aug. 2022 – Jun. 2024

- **I2I: Initializing Adapters with Improvised Knowledge.** Trained CLIP-BART for a method that starts each new task's adapter from a distilled blend of earlier ones. On five VQA tasks it raised new-task scores 4% relative to independent adapters, while AdapterFusion lowered them 2%: reuse works best as a starting point. [7]

Publications

Conference Publications *Equal contribution.

[1] **Batch-of-Thought: Cross-Instance Learning for Enhanced LLM Reasoning**
Xuan Yang, **Furong Jia**, Roy Xie, Xiong Xi, Hengwei Bian, Jian Li, Monica Agrawal.
EMNLP 2026 Main

[2] **Counting Clues: A Lightweight Probabilistic Baseline Can Match an LLM**
Furong Jia*, Yuan Pu*, Finn Guo, Monica Agrawal.
AACL-IJCNLP 2026 Main

[3] **Interpreting Dataset Shift in Clinical Notes**
Shariar Vaez-Ghaemi*, **Furong Jia***, Monica Agrawal.
ML4H 2025. PMLR 297:243–262, 2026.

[4] **How does my language model understand clinical text?**
Furong Jia, David Sontag, Monica Agrawal.
CHIL 2025. PMLR 287:720–743.

[5] **GPT4MTS: Prompt-Based Large Language Model for Multimodal Time-Series Forecasting**
Furong Jia, Kevin Wang, Yixiang Zheng, Defu Cao, Yan Liu.
EAAI 2024. Proc. AAAI 38(21):23343–23351.

[6] **TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting**
Defu Cao, **Furong Jia**, Sercan Ö. Arık, Tomas Pfister, Yixiang Zheng, Wen Ye, Yan Liu.
ICLR 2024.

[7] **I2I: Initializing Adapters with Improvised Knowledge**
Tejas Srinivasan, **Furong Jia**, Mohammad Rostami, Jesse Thomason.
CoLLAs 2023. PMLR 232:923–935.

Workshop / Findings & Abstracts

[8] **What Patients Really Ask: Exploring the Effect of False Assumptions in Patient Information Seeking**
Raymond M. Xiong, **Furong Jia**, Lionel Wong, Monica Agrawal.
ML4H 2025 Findings. arXiv:2601.15674, 2026.

[9] **Leveraging Large Language Models to Identify Barriers to Kidney Transplant in the STEPS Study**
Monica Agrawal, Metin Gurcan, Sarah Peskoe, Freya Gulamali, **Furong Jia**, Tara Strigo, Ebony Boulware, Clarissa Diamantidis.
Am. J. Kidney Dis. 87(5):S203–S204, 2026. (Abstract)

Preprints

[10] **Challenges of Auditing: Variability in Outputs of Large Language Models for Health**
Yuan Pu, Yewon Chang, **Furong Jia**, Xunjian Yin, Jessica Ma, Ayman Ali, Monica Agrawal.
arXiv:2609.16590, 2026.

Manuscripts Under Review

[11] **How Faithfully do Language Models Acquire Probabilistic Knowledge?**
Furong Jia, Xunjian Yin, Yuan Pu, Monica Agrawal.

[12] **When Does the Harness Matter for LLM-Driven Discovery?**
Furong Jia, Xinheng Ding, Chengrui Qu, Monica Agrawal, Akash Srivastava.

Honors and Awards

NeurIPS: Top Reviewer (2026).

University of Southern California: Albert Dorman Future Leader Award (2024); Student Recognition Awards (2024); Provost's Research Fellowship (Fall 2023); CURVE Research Fellowship (2022–2023); Academic Achievement Award (2021); ABC Innovation Prize (2021).

Service

- **Reviewer:** NeurIPS (2025, 2026); ICLR (2027); ML4H (2025, 2026); ACL ARR: Feb./Jun./Oct. 2024, Feb./May 2025; NeurIPS FMTS Workshop (2026); COLM DAIH Workshop (2026).
- **Teaching Assistant, Duke:** Natural Language Processing (Spring 2026); Applied Machine Learning (Spring 2025).
- **Teaching Assistant, USC:** Algorithms and Theory of Computing (Spring 2022); Web Publishing and Front-end Development (Fall 2021 – Spring 2024).